The International Amateur Radio Union

Since 1925, the Federation of National Amateur Radio Societies
Representing the Interests of Two-Way Amateur Radio Communication

AMATEUR SATELLITE FREQUENCY COORDINATION REQUEST INSTRUCTIONS AND APPLICATION FORM[1]

1. Amateur-satellite service. Amateur stations meet the requirements of the radio regulations[2], RR 1.56. and 1.57.

> **RR 1.56** *amateur service:* A *radiocommunication service* for the purpose of self-training, intercommunication and technical investigations carried out by amateurs, that is, by duly authorized **[licensed]** persons **[individual natural people]** interested in radio technique solely with a personal aim **[for themselves]** and without pecuniary interest **[compensation]**. (NOTE: Explanatory terms in brackets are not part of the treaty text.)
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> **RR 1.57** *amateur-satellite service:* A *radiocommunication service* using *space stations* on earth *satellites* for the same purposes as those of the *amateur service*.

Before asking for help from IARU with frequency coordination in the amateur-satellite service, make sure that your proposed operation meets the treaty requirements. NOTE: "Without pecuniary interest" means that you may accept free will donations of goods and services, that is, with nothing required in return. You may not sell services or data to anyone for any reason.

Ultimately, the decision of whether the proposed operation is appropriate for the amateur-satellite service rests with your country's administration (your national telecommunication regulator). Therefore, before sending your frequency coordination request to IARU, we suggest that you consult with your administration to determine whether the amateur-satellite service or another radiocommunication service is appropriate for your operation.

2. Self-coordination. For over 100 years, amateur radio operators have maintained an effective tradition of self-regulation. Amateurs are expected to coordinate their use of frequencies. (Nobody has any exclusive right to use any particular frequency.) Coordination of many terrestrial stations, repeaters and beacons, for example, usually works well through IARU member national societies and local coordinating committees.

Coordinating satellites. Amateur radio satellites present a special problem because satellites have global effect. Only a global frequency coordination system can work. Uncoordinated satellites will cause harmful interference to stations around the world and receive interference from them — which could result in mission failure.

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[1] Terms used here are defined in the radio regulations and repeated in the IARU paper, *Amateur Satellites*. See:. http://www.iaru.org/uploads/1/3/0/7/13073366/iarusatspec_rev15.7.pdf

[2] The radio regulations are annexed to and part of a treaty, the International Telecommunication Convention, to which nearly every country in the world is a signatory party. See: http://www.itu.int/pub/S-CONF-PLEN-2011.

Coordination serves everyone's best interests!

3. IARU Coordination procedure.

a) Frequency coordination for amateur radio satellites is provided by the IARU through its Satellite Advisor, a senior official appointed by the IARU Administrative Council, its top policymaking body. The IARU Satellite Advisor is assisted by an Advisory Panel of qualified amateurs from all three IARU Regions. (Similar to ITU Regions.)

b) It is a mandatory requirement for the IARU coordination that the licensing administration notifies the ITU with the selected frequency assignments/bands for the proposed satellite (ref RR Article 9, sub-section IA[3]). The licensing administration files the Advanced Publication Information (API) with ITU. This should preferably be done prior to the actual frequency coordination by IARU. As a response to this submission, the ITU will assign a special section number (API/A). This API/A special section number must be provided in field 1d of the Frequency Coordination Request form. If this information is not available when the IARU Frequency Coordination Request is filed, please provide the date that you submitted the API information to your administration. Usually an API special section publication appears within two to three months after its submission to ITU. It is important that you encourage your national administration to forward their notice to ITU as early as possible.

Note that for the amateur satellite service, ITU offers this registration without any fee of cost recovery.

c) It is important that all satellites operating in the amateur satellite service are registered with ITU in order to obtain international recognition, something that demonstrates the relevance of the amateur satellite service and protects its frequency allocations. IARU therefore strongly recommends that you work with your national administration and encourage them after the API submission to continue with the submission of the Notification form of notice of the satellite stations when they are brought into use (RR Article 11.2). Assistance with the notification process is provided on the ITU web.

4. Getting Help.

a) **Start** by reading *Amateur Radio Satellites*, which is an IARU paper. You will find explanations and interpretations of Treaty provisions. IARU satellite frequency coordination follows these interpretations. Download the latest version from: https://www.iaru.org/wp-content/uploads/2019/12/iarusatspec_rev15.7.pdf

b) **Discuss** your project with the national amateur radio society of your country and your national AMSAT organization, if there is one. They may be able to assist you in a variety of ways.

c) **Use information** available on-line.

i. For a list of national amateur radio societies (Member Societies of IARU), see: http://www.iaru.org/iaru-soc.html.

[3] This applies to non-GSO amateur satellites only. For GSO satellites contact the IARU Satellite Advisor for further guidance.

ii. For a list of amateur satellite organizations, see: https://www.amsat.org/amsats-around-the-world/

iii. A link budget spread sheet is at: http://www.amsatuk.me.uk/iaru/spreadsheet.htm

iv. Check frequencies of currently operating satellites at https://www.amsat.org/status/
Check on coordinated and other planned amateur satellites at: http://www.amsat.org.uk/iaru/.

v. If you need help understanding the requirements or completing the coordination request, ask the Satellite Advisor or a Panel Member.

5. When to make the frequency coordination request. Make your frequency coordination request as far in advance as possible. Remember, coordination takes account of your own needs and the needs of others. Receiving coordination early enough makes design and construction easier and less expensive. In any event, be sure to make your request while it is still possible to change operating frequencies in response to the Satellite Advisor's recommendations. A good moment to submit your coordination request would be right after you have done your preliminary Design Review.

6. Who makes the frequency coordination request? The prospective space station licensee must make the coordination request, as that person will be responsible for space station transmitter operations.

7. Where to send your frequency coordination request. Send frequency coordination requests to the IARU Satellite Advisor by e-mail to satcoord@iaru.org. You should get an acknowledgement that the request has been received within a week.

8. What will happen? The IARU Satellite Advisor will make recommendations to the licensee concerning plans based upon all available information and advice from the Satellite Advisory Panel. His goal is to help you and your project to succeed. The current status of your request will be published through the link at http://www.amsat.org.uk/iaru. When the process is complete, the licensee will receive a coordination letter with detailed information. Visit also http://www.iaru.org for information about the IARU and its activities.

VERY IMPORTANT!

CHECKLIST:

1. ☐ **USE THE FREQUENCY COORDINATION REQUEST FORM PROVIDED IN THE PAGES BELOW.**

2. ☐ **NAME THE ELECTRONIC DOCUMENT** you submit with the name of the proposed satellite followed by the submission date. Example: if the name before launch is NewsatA and the document is submitted on 1 August 2017, the document file name should be: "newsata_req 170801.doc."

3. ☐ **LARGE FILES** should **NOT** be in the request form. **INDICATE** URL's for pictures, sketches, drawings, and other pertinent information.

4. ☐ **INDICATE** in box number 13 whether or not you feel that the proposed operation in the amateur-satellite service is consistent with the radio regulations as interpreted by the IARU Satellite Advisor. If not, please, explain your interpretation of the radio regulations. Tic only ONE box.

5. ☐ **REMOVE** RED explanatory texts in the form below.

6. ☐ **LICENSEE** must sign and date the form in boxes 3.8, 6.4 and 14.

7. ☐ **SUBMIT THE COMPLETED FORM** as soon as possible, **do NOT submit the request before it is 100% complete and signed**. Incomplete and unsigned requests will NOT be assessed by the IARU frequency coordination panel.

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AMATEUR SATELLITE FREQUENCY COORDINATION REQUEST

(Make a separate request for each space station to be operated in the amateur-satellite service.)

	Administrative information and document control	
1.1	Date of this submission	06-AUG-2026
1.2	Submission revision number	0

	Spacecraft identification	
2.1	Name	LEO-Gamma Ray Mapping
2.2	Notifying administration	FCC (United States of America)
2.3	API/A number	"Pending Submission", will be updated automatically upon filing by the administration (FCC).

	Licensee of the space station or responsible amateur in case of educational mission	
3.1	First (given) name?	Will
3.2	Last (family) name?	Farr
3.3	Amateur radio call sign	Licensee's own station call sign.
3.4	Licensed since?	When you obtained your license.
3.5	Postal address?	Stony Brook University, Department of Physics and Astronomy, 100 Nicolls Road, Stony Brook, NY 11794
3.6	Telephone number (including country code)?	
3.7	E-mail address?	will.farr@stonybrook.edu
3.8	By signing here, the licensee accepts that, except for the	Signature:

	telecommand information, the IARU may publish all the information contained in this application.	Acknowledgment that the given information (except telecommand) in the application will be published. Date:

	Organizations — complete this section for EACH participating organization	
4.1	Name of organization and/or educational institution?	Stony Brook University
4.2	Licensee's position in any organization referenced in item 4.1?	Faculty Advisor and Project Trustee
4.3	Postal address?	Postal address of organization if different to that given in 3.5.
4.4	Telephone number? (including country code)	
4.5	E-mail address?	will.farr@stonybrook.edu
4.6	Web site URL?	https://leosgxsbu.github.io/LEOSGX-1UCubeSat/
4.7	National Amateur Radio Society (including contact information)?	AARL (American Radio Relay League) 224 Main Stret, Newington, CT 0611
4.8	Does your National Amateur Satellite organization and/or National Amateur Radio Society endorse this request?	National AMSAT Organization ☐yes ☐no National Amateur Radio Society ☐yes ☐no
4.9	Name and email address(es) of the person(s) you've contacted in the National AMSAT Organization or National Amateur Radio Society?	Name: Email: Name and contact details of society personnel contacted in relation to this request. Please attach any documents from the societies that support your mission proposal
4.10	Will any person or organization involved with the project be, directly or indirectly, financially compensated for operating the satellite and ground station(s)?	☐yes ☐no

4.11	If you've answered yes in 4.10, please explain:	Answer was no.

	Space station mission and frequencies	
5.1	Type of mission? Tick applicable box(es):	☐Amateur ☐Amateur combined with Educational ☐Amateur combined with other mission(s)
5.2	If mission type in given in 5.1 is Amateur combined with other missions, will the transmitters or receivers operating in the amateur radio frequencies be used to control, or retrieve data (telemetry, payload, etc.) from, the non-amateur mission sub-systems?	☐yes ☐no
5.3	If you've answered yes in 5.2, please explain:	Answer was no
5.4	List and describe the project mission(s):	The LEO-Gamma Ray Mapping is an educational 1U CubeSat astrophysics mission designed and manufactured by an undergraduate team at Stony Brook University. The satellite will be carrying a Thallium-doped Cesium Iodide [Csi(TI)] crystal plate coupled to a Silicon Photomultiplier array. The main objective is to record data from cosmic events, specifically mapping the fluence and tiny duration profiles of short Gamma-Ray Bursts (GRBs).
5.5	Explain in plain text how your mission(s) complies with provisions no. **1.56**, **1.57** and **25** (see Annex 1); how will your mission contribute to the advancement of the amateur satellite service; and how will amateur operators around the globe be able to participate in your mission, besides just receiving the satellite telemetry?	The spacecraft is designed and will operate without any commercial intent. The whole codebase, digital files, mechanical files, the telemetry data will all be open-sourced and published freely on our project repository. Radio amateurs globally can openly participate by tuning into the 437 MHz frequency band and downloading the telemetry data, decentralized open-source archives via the global SatNOGS tracking ground grid.
5.6	Amateur satellite bands used:	☐144-146 MHz ☐435-438 MHz ☐1260-1270 MHz

		☐2400-2450 MHz ☐3400-3410 MHz ☐5650-5670 MHz and/or 5830-5850 MHz ☐10.450-10.500 GHz ☐24.000-24.050 GHz ☐Other (please list)
5.7	Planned duration of each part of the mission?	Phase 1 : For 45 minutes, there will be a delay in the power delivery to all stages and all transmission is silent. Phase 2 : The antennas are deployed. Phase 3 : It is going to be receiving commands and sending binary data back and has an expected life of around 1 to 2 years.
	Proposed space station transmitting (space-to-Earth and/or space-to-space) frequency plan. Provide the following information for each frequency or frequency range:	
5.8	List all the frequencies (or frequency bands) requested and describe the function of each:	A single channel within the 435-438 MHz space allocation. This channel will act as a double pathway for sending binary data packets from the satellite and commands from the University to the Satellite.
5.9	Frequency tuning range (in MHz) of transmitter and tuning step increment (in Hz or kHz)?	420.00MHz — 450.00MHz 1.0 kHz step resolution
5.10	E.I.R.P (in dBm) of each transmitter?	+20 dBm (100 mW) – Single Transmitter
5.11	List all ITU emission designator for each transmitter:	14K0G1D
5.12	Common description of the emissions including modulation type AND data rate?	9600 Baud packet Radio utilizing GFSK Modulation and AX.25 standard packet framing.
5.13	Type of antenna, antenna gain and radiation pattern?	Deployable two 16.3 cm flexible steel tape-measure radiator elements emitting a single direction radiation pattern.
	Proposed space station receiving (Earth-to-space and/or space-to-space) frequency plan. Provide the following information for each frequency or frequency range:	

5.14	Requested receive frequency and function?	435.00 MHz to 438.00 MHz
5.15	Frequency tuning range (in MHz) of receiver and tuning step increment (in Hz or kHz)?	420 MHz to 450 MHz 1 kHz step resolution
5.16	ITU emission designator for each transmitter and emission type?	14K0G1D
5.17	Common description of the emission including modulation type AND data rate for each transmitter and emission type?	9600 Baud GFSK AX.25 command packet streams.
5.18	Noise temperature for each receiver onboard the satellite?	Around 290 Kelvins
5.19	Associated antenna gain and radiation pattern for each receiver onboard the satellite?	Antenna Shared Dipole interface. Peak gain 2.15 dBi
	Other information	
5.22	Physical structure of the space station, including dimensions and mass?	Standardized 1U CubeSat Mass : 1.33 kg Dimensions : 100.0mm X 100.0mm X 113.5mm
5.23	Functional Description of each satellite sub-system, including non-amateur payloads?	Six-layer integrated internal vertical PCB stack on a unified 95x95mm layout profile. Tier 1 holds the CsI(Tl) scintillation crystal and Hamamatsu SiPM photon capture array. Tier 2 contains the analog Pico Spectrometer noise filtering loop. Tier 3 is the STM32F411C computer processor hub. Tier 4 handles the solar power battery charging networks. Tier 5 acts as the structural cell floor ballast hosting our transceiver module, and Tier 6 functions as the antenna release frame
5.24	Electrical Power budget. (average power consumption and power generation in Watts)?	Average Power Generation : 1.5 W Average Power Consumption : 0.8 W
5.25	Method of attitude stabilization (if used)?	Passive Magnetic Attitude Control (PMAC). Dual permanent Z–axis Neodymium alignment bar magnets

5.26	Service area?	Intended service area of satellite if not worldwide.

	Encryption of downlink?	
6.1	Will all transmissions from the satellite have published and publicly available details on the project web site? (Note: The signal **must not** be effectively encrypted by not providing the necessary information. See No. **25.2A** in Annex 1)	☐yes ☐no, see 6.2
6.2	If you've answered no in 6.1, please explain:	Answer was no
6.3	Contact details of the person responsible for publishing the descriptions of modulation, protocol, format, etc.?	Muhammad Maalik and Professor Will Farr muhammad.maalik@stonybrook.edu will.farr@stonybrook.edu
6.4	By signing here, the responsible person listed in 6.3, agrees that the details required in 6.1 will be made publicly and freely available.	Signature: Muhammad Maalik & Date:8/20/2026

	Telecommand Information (NOT published)	
7.1	Telecommand station amateur radio callsign?	List telecommand station call signs.
7.2	Physical location of station antennas, including latitude and longitude?	List physical locations of each telecommand station
7.3	Proposed space station telecommand frequencies?	435MHz – 438MHz
7.4	List all ITU emission designators for each transmitter:	14K0G1D

7.5	Common description of the emission including modulation type AND data rate?	9600 Baud GFSK command streams inside standard AX.25 framework boundaries.
7.6	Radio Link budget(s):	Link budget calculations are currently under calculation. The finalized tracking balance sheet will be hosted openly on the public project repository directory
7.7	A general description of any telecommand encryption:	The satellite transmits open source telemetry data using the standard amateur AX.25 packet format at 9600 Baud. All data parsing dictionaries and hexadecimal decoding equations will be hosted publicly on our project repository website listed in Field 4.6, allowing radio amateurs worldwide to decode the science data packages.
7.8	Explain how telecommand stations will turn off the space station transmitter(s) immediately, even in the presence of user traffic and/or space station computer system failure. See No. **22.1** in Annex 1.	The telecommand node uses a hardwired logic override path on our internal bus circuit. When an emergency transmission command is sent via the uplink receiver, this will bypass all flight computer main loop software and asserts a hard logic trigger directly cutting the VCC Power Trail supplying the RFM95W radio module. This guarantees immediate cessation of emissions under the RR Article 22.1
7.9	Explain the mechanism to ensure the space station is turned off in the event communications is lost with the command station.	The flight computer firmware operates an independent background software timer. If the satellite does not successfully log and clear a command ping from the tracking station withing 72 hours/ The software executes a permanent lockout, cutting power to the radio and doing a recovery boot.

	Telemetry frequencies	
8.1	All amateur telemetry frequencies or frequency bands used in the mission?	Shared single-channel half-duplex operational slot inside the requested 435-438 MHz band
8.2	ITU emission designators:	14K0G1D
8.3	Common description of the emission including modulation type AND data rate?	9600 Baud GFSK AX.25 telemetry beacon transmissions
8.4	Radio Link budgets:	Link budget calculations are currently under calculation. The finalized tracking balance sheet will be hosted openly on the public project repository directory
8.5	Transmission formats e.g. AX25 etc.? Provide any other	The satellite transmits open source telemetry data using the standard amateur AX.25 packet format at 9600 Baud.

	details required to understand the telemetry transmission:	All data parsing dictionaries and hexadecimal decoding equations will be hosted publicly on our project repository website listed in Field 4.6, allowing radio amateurs worldwide to decode the science data packages.

	Launch plans	
9.1	Launch agency?	Pending official NASA launch assignment
9.2	Launch location?	Pending official NASA launch assignment
9.3	Expected launch date?	Pending official NASA launch assignment
9.4	Planned orbit apogee?	Pending official NASA launch assignment
9.5	Planned orbit perigee	Pending official NASA launch assignment
9.6	Planned orbit inclination	Pending official NASA launch assignment
9.7	Planned orbit period	Pending official NASA launch assignment
9.8	List other amateur satellites expected to share the same launch (if known):	Pending official NASA launch assignment

	Typical Earth station — transmitting	
10.1	Describe the hardware and software of a typical Earth station to transmit signals to the planned satellite:	The downlink tracking network will rely on the SatNOGS software. Raw captured frames will be posted straight to the public database archives over Wifi in real time.
10.2	Radio Link budget. Show complete link budgets for all Earth station transmitting frequencies, except telecommand.	Not applicable. The Earth ground station terminal does not transmit any amateur space frequencies on this mission, except for the secure, half-duplex telecommand signals already defined and listed in Section 7.

	Typical Earth station — receiving	
11.1	Describe the hardware and software of a typical Earth station to receive signals from	The ground receiving station will utilize standard, opensource SatNOGS tracking terminals equipped with a UHF antenna and a Software Defined Radio (SDR)

	the planned satellite.	receiver to automatically capture and decode the satellite's downlink packets.
11.2	Radio Link budget. Show complete link budgets for all Earth station receiving frequencies.	Link budget calculations are currently under calculation. The finalized tracking balance sheet will be hosted openly on the public project repository directory

	Additional information: Do not attach large files. Indicate the URL where the information is available.
12	Please, supply any additional information that may assist the Satellite Advisor to coordinate your request(s).

	Certification: **Please tick ONE appropriate box.**
13	☐The licensee of the planned space station has reviewed all relevant laws, rules, and regulations, and certifies that this request complies with all requirements as understood by IARU to the best of his/her knowledge and confirms to meet the requirements of RR 1.56 and RR 1.57 in that the proposed satellite will operate without pecuniary interest. **Please list any commercial interests. If none, please state none.** None`
	☐The licensee of the planned space station has reviewed all relevant laws, rules, and regulations and disagrees with IARU interpretations of Treaty requirements. The IARU Satellite Advisor is asked to consider the following interpretation. Explanation follows.

	Signature	
14	Signature of licensee responsible for this application and operation of the satellite system Signature of space station licensee	Date

Document History
29/4/2020: Revision of V39

Annex 1: Extracts from the ITU Radio Regulations

1.56 *amateur service:* A *radiocommunication service* for the purpose of self-training, intercommunication and technical investigations carried out by amateurs, that is, by duly authorized persons interested in radio technique solely with a personal aim and without pecuniary interest.

1.57 *amateur-satellite service:* A *radiocommunication service* using *space stations* on earth *satellites* for the same purposes as those of the *amateur service*.

ARTICLE 25

Amateur services

Section I − Amateur service

25.1 § 1 Radiocommunication between amateur stations of different countries shall be permitted unless the administration of one of the countries concerned has notified that it objects to such radiocommunications. (WRC-03)

25.2 § 2 1) Transmissions between amateur stations of different countries shall be limited to communications incidental to the purposes of the amateur service, as defined in No. **1.56** and to remarks of a personal character. (WRC-03)

25.2A 1A) Transmissions between amateur stations of different countries shall not be encoded for the purpose of obscuring their meaning, except for control signals exchanged between earth command stations and space stations in the amateur-satellite service. (WRC-03)

25.3 2) Amateur stations may be used for transmitting international communications on behalf of third parties only in case of emergencies or disaster relief. An administration may determine the applicability of this provision to amateur stations under its jurisdiction. (WRC-03)

25.4 (SUP - WRC-03)

25.5 § 3 1) Administrations shall determine whether or not a person seeking a licence to operate an amateur station shall demonstrate the ability to send and receive texts in Morse code signals. (WRC-03)

25.6 2) Administrations shall verify the operational and technical qualifications of any person wishing to operate an amateur station. Guidance for standards of competence may be found in the most recent version of Recommendation ITU-R M.1544. (WRC-03)

25.7 § 4 The maximum power of amateur stations shall be fixed by the administrations concerned. (WRC-03)

25.8 § 5 1) All pertinent Articles and provisions of the Constitution, the Convention and of these Regulations shall apply to amateur stations. (WRC-03)

25.9 2) During the course of their transmissions, amateur stations shall transmit their call sign at short intervals.

25.9A § 5A Administrations are encouraged to take the necessary steps to allow amateur stations to prepare for and meet communication needs in support of disaster relief. (WRC-03)

25.9B § 5B An administration may determine whether or not to permit a person who has been granted a licence to operate an amateur station by another administration to operate an amateur station while that person is temporarily in its territory, subject to such conditions or restrictions it may impose. (WRC-03)

Section II − Amateur-satellite service

25.10 § 6 The provisions of Section I of this Article shall apply equally, as appropriate, to the amateur-satellite service.

25.11 § 7 Administrations authorizing space stations in the amateur-satellite service shall ensure that sufficient earth command stations are established before launch to ensure that any harmful interference caused by emissions from a station in the amateur-satellite service can be terminated immediately (see No. **22.1**). (WRC-03)

ARTICLE 22

Space services[1]

Section I − Cessation of emissions

22.1 § 1 Space stations shall be fitted with devices to ensure immediate cessation of their radio emissions by telecommand, whenever such cessation is required under the provisions of these Regulations.

[1] **A.22.1** In applying the provisions of this Article, the level of accepted interference (see No. **1.168**) shall be fixed by agreement between the administrations concerned, using the relevant ITU-R Recommendations as a guide.